

Front-Arm-Subsystem-PowerBudget

Team Number: 202

Project Name: S.C.O.U.T.S

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Version: 1

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Stepper Motor		+10.8V	1	2500	2500	mA
	DRV8434S (motor driver)	DRV8434S	3.3V logic / VM	1	10	10	mA
	ESP32-S3-WROOM-1	ESP32-S3-WROOM	3.3V	1	500	500	mA
	LM22678 (5V regulator)	LM22678	4.5-42V	1	1000	1000	mA
	LM2651 (3.3V regulator)	LM2651	4-14V	1	1500	1500	mA
+10.8V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Stepper Motor		+10.8 V	1	2500	2500	mA
	LM22678 input	LM22678	4.5-42V	1	700	700	mA
						0	mA
	Subtotal					3200	mA
	Safety Margin					25%	
	Total Current Required on +10.8V Rail					4000	mA
Regulator	Battery Pack	BL5000C21703S2P	10.8V nominal	1	8000	8000	mA
	Total Remaining Current Available on +10.8V Rail					4000	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	LM2651 (3.3V regulator)	LM2651	4-14V	1	600	600	mA
							mA
						0	mA
			Subtotal			600	mA
			Safety Margin			25%	
			Total Current Required on +5V Rail			750	mA
Regulator	+5V Regulator	LM22678	4.5-42V	1	5000	5000	mA
	Total Remaining Current Available on +5V Rail					4250	mA

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	DRV8434S (motor driver)	DRV8434S	3.3 V- VM	1	10	10	mA
	ESP32-S3-WROOM-1	ESP32-S3-WROOM-1	3.3V	1	500	500	mA
	Subtotal					510	mA
	Safety Margin					25%	
	Total Current Required on +3.3V Rail					637.5	mA
Regulator	+3.3V Regulator	LM2651	4-14V	1	1500	1500	mA
	Total Remaining Current Available on 3.3V Rail					862.5	mA

<i>External Power Source 1</i>	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Battery Pack	Battery Pack	BL5000C21703S2PF	10.8 V nominal	10.8V	8000	8000	mA
Total Remaining Current Available on External Power Source 1						8000	mA

<i>External Power Source 2</i>	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Total Remaining Current Available on External Power Source 2						0	mA

Calculated Battery Life							
	Component Name	Part Number	Supply Voltage Range		Capacity (mAh)	Required By Regulators	
	Battery Pack	BL5000C21703S2PF	10.8 V nominal	10.8V	10000	4000	
Battery Life						2.5	hours

Notes:
 External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
 If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit